MGT-424: Machine Learning Methods in Econometrics

Homework 2

Question 1: Bayesian Networks and Chow–Liu Trees

In this homework, we study probabilistic graphical models, conditional independence, and structure learning using the Chow–Liu algorithm.

Suppose we observe four binary random variables

$$X_1, X_2, X_3, X_4$$

and the empirical pairwise mutual informations between them are given below.

Pair	Mutual Information
$I(X_1; X_2)$	0.82
$I(X_1; X_3)$	0.10
$I(X_1; X_4)$	0.41
$I(X_2; X_3)$	0.51
$I(X_2; X_4)$	0.33
$I(X_3; X_4)$	0.75

We wish to approximate the unknown joint distribution

$$P(X_1, X_2, X_3, X_4)$$

using a tree-structured Bayesian network.

(i) **Constructing the Chow–Liu Tree**

(a) Construct the weighted graph whose edge weights are the mutual informations above.

(b) Find the maximum-weight spanning tree.

(c) Draw the resulting tree.

(ii) **From Tree to Bayesian Network**

Choose any node as a root and orient all edges away from it to obtain a Bayesian network.

Write the corresponding factorization of the joint distribution.

(iii) **Conditional Independence**

Using your graph:

(a) State one pair of variables that are conditionally independent given another variable.

(b) Verify your answer using d-separation.

(c) Explain intuitively why this conditional independence holds.

(iv) **Why Does Chow–Liu Work?**

The Chow–Liu theorem states that the optimal tree approximation minimizes

$$D(P\|Q)$$

among all tree-structured distributions Q.

Explain intuitively why maximizing the sum of pairwise mutual informations leads to a good approximation of the true distribution.

You do not need to provide a formal proof.

(v) **Programming Part: Chow–Liu from Scratch and Verification**

In this part, you will implement the Chow–Liu algorithm.

You are **not allowed** to use graph libraries such as `networkx`, `igraph`, or any built-in function for computing spanning trees in Part (v.i).

Part (v.i): Implementation from Scratch

Implement Kruskal's algorithm from scratch to compute the maximum-weight spanning tree.

Complete the missing parts in the following code.

```python
variables = ["X1", "X2", "X3", "X4"]

edges = [
    ("X1", "X2", 0.82),
    ("X1", "X3", 0.10),
    ("X1", "X4", 0.41),
    ("X2", "X3", 0.51),
    ("X2", "X4", 0.33),
    ("X3", "X4", 0.75)
]

def find(parent, x):
    """
    Return the representative of the set containing x.
    """
    # TODO: Implement the find operation.
    pass

def union(parent, rank, x, y):
    """
    Merge the sets containing x and y.
    """
```

```python
    # TODO: Implement union by rank.
    pass

def chow_liu_tree_from_scratch(variables, edges):
    """
    Computes the maximum-weight spanning tree using Kruskal's
        algorithm.
    """
    parent = {}
    rank = {}

    # TODO: Initialize parent and rank for each variable.

    # TODO: Sort edges in decreasing order of mutual information.
    sorted_edges = None

    tree_edges = []

    for u, v, w in sorted_edges:
        # TODO: Check whether u and v are already connected.

        # TODO: If not connected, add the edge to tree_edges
        # and merge the two components.

        # TODO: Stop when the tree has len(variables) - 1 edges.
        pass

    return tree_edges

tree = chow_liu_tree_from_scratch(variables, edges)

print("Edges in the Chow--Liu tree from scratch:")
for u, v, w in tree:
    print(u, "--", v, "weight =", w)
```

Report the selected edges and the total weight of the tree.

Part (v.ii): Verification with `networkx`

Now use `networkx` only to verify your answer from Part (v.i).

Complete the code below and compare the result with your implementation.

```python
import networkx as nx

G = nx.Graph()

# TODO: Add the weighted edges to G.

# TODO: Compute the maximum spanning tree using networkx.
T_nx = None

print("Edges in the Chow--Liu tree using networkx:")
# TODO: Print the edges of T_nx with their weights.
```

```
# TODO: Compare the set of edges from your implementation
# with the set of edges obtained from networkx.
```

Answer the following questions:

(a) Do both implementations return the same tree?

(b) If there are multiple maximum spanning trees with the same total weight, what kind of differences could appear?

(c) Why is it important to check for cycles when constructing the tree?

(vi) **Limitation of Tree Models**

Can every probability distribution be represented exactly by a tree-structured Bayesian network?

Briefly explain your answer.

Question 2: Potential Outcomes, Causal Graphs, and SEM Using the LaLonde Dataset

Important coding rule: In this exercise, you should only use `numpy` and `pandas` for the Python code. Do not use `statsmodels`, `sklearn`, or other regression packages.

In this exercise, we use the LaLonde job training dataset. The dataset can be downloaded from:

`https://raw.githubusercontent.com/robjellis/lalonde/master/lalonde_data.csv`

The dataset is used to study whether participation in a job training program affected workers' real earnings in 1978.

The important variables are:

- `treat`: equals 1 if the person received job training, and 0 otherwise.
- `re78`: real earnings in 1978.
- `age`: age of the person.
- `educ`: years of education.
- `black`: equals 1 if the person is African-American, and 0 otherwise.
- `hispan`: equals 1 if the person is Hispanic, and 0 otherwise.
- `married`: equals 1 if the person is married, and 0 otherwise.
- `nodegree`: equals 1 if the person has no high school degree, and 0 otherwise.
- `re74`: real earnings in 1974.
- `re75`: real earnings in 1975.

Questions

1. Identify the treatment variable and the outcome variable in this dataset. Explain in words what each variable means in this study.

2. Define the potential outcomes $Y_i(1)$ and $Y_i(0)$ for worker i, and write the individual treatment effect using these potential outcomes.

3. Explain why we cannot observe both $Y_i(1)$ and $Y_i(0)$ for the same worker.

4. Define the Average Treatment Effect (ATE) using potential outcome notation.

5. Estimate the naive difference in means:

$$\widehat{ATE}_{naive} = \bar{Y}_{treated} - \bar{Y}_{control}$$

   where $Y =$ `re78`.

6. Interpret the sign and size of the naive estimate. Does the naive estimate suggest that the job training program increased or decreased earnings?

7. Why might the naive difference in means fail to estimate the true causal effect of job training? Provide an example using variables from this dataset.

8. Create a guessed causal graph using all of the variables in the dataset:

$$treat,\ age,\ educ,\ black,\ hispan,\ married,\ nodegree,\ re74,\ re75,\ re78.$$

   Your graph does not have to be uniquely correct. The goal is to make a reasonable graph and explain your reasoning.

9. Make a Structural Equation Model (SEM) using linear models for both treatment and outcome. Use all relevant pre-treatment variables in the treatment equation and the outcome equation.

   Specifically, write one linear model for treatment participation based of all other variable except the outcome, then write another linear model for the outcome based on all other variables.

10. Use ordinary least squares regression (OLS) to estimate the outcome equation from the SEM in Question 9. Use only `numpy` and `pandas`. What does β_1 represent?

11. Compare the naive estimate and the regression-adjusted OLS estimate. Why might they be different?

12. Does a positive regression-adjusted coefficient prove that job training caused higher earnings? Explain carefully.

Question 3: Unsupervised Learning for Worker Segmentation Using the LaLonde Dataset

Important coding rule: For the *main implementation* in this question, you should use only `numpy`, `pandas`, and `matplotlib`. You may use the `KMeans` and `PCA` implementations from `sklearn` **only in the final verification part**.

Econometric datasets often contain substantial heterogeneity across individuals, households, firms, or regions. Unsupervised learning methods can help summarize this heterogeneity, detect latent groups, and construct low-dimensional representations before a formal causal or predictive analysis.

In this exercise, use the **same LaLonde dataset** as in Question 2. Your goal is to uncover groups of workers with similar pre-treatment characteristics and to interpret these groups economically.

Use the following **pre-treatment variables** for the unsupervised analysis:

`age`, `educ`, `black`, `hispan`, `married`, `nodegree`, `re74`, `re75`.

Do not use `treat` or `re78` when fitting the unsupervised models. You may use them *only after clustering* for interpretation.

Questions

1. Explain briefly why `treat` and `re78` should be excluded when the purpose is to discover worker types based only on pre-treatment characteristics.

2. Standardize the eight pre-treatment variables using

$$x^* = \frac{x - \bar{x}}{s_x}.$$

   Explain why standardization is important before applying K-means and PCA in this dataset.

3. Let $X_i \in \mathbb{R}^8$ denote the standardized pre-treatment covariates for worker i. Write the K-means objective function for K clusters:

$$\min_{\{C_1,\dots,C_K\},\ \{\mu_1,\dots,\mu_K\}} \sum_{k=1}^{K} \sum_{i \in C_k} \|X_i - \mu_k\|^2.$$

   Explain in words what this objective is trying to do.

4. Implement **K-means from scratch** using Euclidean distance. Your implementation should:
   (a) initialize centroids by randomly selecting observations,
   (b) alternate between assignment and centroid update steps,
   (c) stop when assignments no longer change or when a maximum number of iterations is reached.

5. Run your K-means implementation for

$$K \in \{2, 3, 4, 5, 6\}$$

   using at least 20 random initializations for each K, and keep the solution with the lowest within-cluster sum of squares.

   Report the final within-cluster sum of squares for each K, and create an elbow plot. Based on this plot and on economic interpretability, choose one preferred number of clusters.

6. For your preferred K, report:

(a) the size of each cluster,
(b) the mean of each pre-treatment variable within each cluster,
(c) the treatment rate $\Pr(\texttt{treat} = 1)$ within each cluster,
(d) the average 1978 earnings `re78` within each cluster.

Interpret the clusters economically. For example, do some clusters look younger, more disadvantaged, or more attached to the labor market than others?

7. Within each cluster, compute the naive treated-minus-control mean difference in `re78`:

$$\bar{Y}_{treated,g} - \bar{Y}_{control,g}.$$

Compare these differences across clusters.

Do they suggest possible treatment-effect heterogeneity? Why should these within-cluster differences still **not** be interpreted as clean causal effects?

8. Now perform **Principal Component Analysis (PCA)** on the same standardized pre-treatment variables.

Compute PCA using the covariance matrix and eigenvalue decomposition (or SVD with `numpy`). Report:

(a) the proportion of variance explained by the first two principal components,
(b) the loadings of the first two principal components,
(c) a scatter plot of observations in the $(PC1, PC2)$ plane, colored by your chosen K-means clusters.

9. Interpret the first two principal components. Which original variables appear to drive $PC1$ and $PC2$? Do the components seem to measure dimensions such as labor-market attachment, socioeconomic disadvantage, or prior earnings history?

10. **Verification with libraries.** Use the `KMeans` and `PCA` implementations from `sklearn` only in this part to verify your main results.

Compare:

(a) your K-means objective value with the one from `sklearn`,
(b) your explained-variance results with the ones from `sklearn`,
(c) your cluster assignments with the library assignments.

If the cluster labels are numerically different, explain why this does not necessarily imply a different clustering.

11. Briefly discuss two ways unsupervised learning can be useful in econometrics, and one important limitation.

Your answer should distinguish clearly between:

(a) using unsupervised learning for *description/exploration*, and
(b) making *causal* claims.